

Day 27 – Confidence and Sentiment Analysis Engine

1. Objective

The Confidence and Sentiment Analysis Engine analyzes observable communication signals from candidate screening responses.

The engine evaluates communication-related signals such as hesitation, response length, pace, sentiment, uncertainty, and consistency to generate a structured confidence analysis.

The confidence score represents observable communication strength and is not intended to be a psychological assessment.

2. Processing Capabilities

The Day 27 engine supports:

- Hesitation pattern detection
- Response length analysis
- Response pace analysis
- Sentiment analysis
- Uncertainty detection
- Consistency analysis
- Confidence score calculation
- Confidence category classification
- Structured analysis output

3. Analysis Flow

Candidate Response | v Text Normalization | v Communication Signal Detection | +-----+-----
-----+ || | v v v Hesitation Sentiment Uncertainty Detection Detection Detection || | +-----
-----+-----+ | v Response Length & Pace | v Consistency Analysis | v Weighted Score
Calculation | v Confidence Score and Category

4. Configuration

The engine uses the following configuration file:

data/confidence_sentiment_configuration.json

The configuration defines:

- Score range
- Signal weights
- Confidence thresholds
- Hesitation terms
- Uncertainty terms
- Positive sentiment terms

- Negative sentiment terms

5. Scoring Criteria

The confidence analysis uses six observable communication criteria.

5.1 Hesitation

Hesitation indicators include conversational expressions such as:

- Um
- Uh
- Er
- Ah
- Hmm
- You know
- Let me think
- I mean

These terms are used as observable speech signals during analysis.

5.2 Response Length

The engine evaluates the length of the candidate response.

Response length is treated as one communication signal and contributes to the overall confidence score according to the configured weight.

5.3 Response Pace

Response pace is included as an observable communication signal.

The engine uses the available response timing information to evaluate the pace of the candidate response.

5.4 Sentiment

The engine identifies basic positive and negative sentiment indicators.

Positive terms include examples such as:

- Confident
- Successfully
- Achieved
- Improved
- Excellent
- Strong
- Enjoy
- Comfortable
- Experienced

Negative terms include examples such as:

- Difficult
- Problem
- Failed
- Failure
- Weak
- Worried
- Confused
- Unable

5.5 Uncertainty

The engine detects uncertainty expressions such as:

- Maybe
- Perhaps
- I think
- I guess
- Not sure
- Probably
- Might be
- I don't know

These expressions are treated as observable uncertainty signals.

5.6 Consistency

The engine evaluates consistency indicators from the available response information.

Consistency contributes to the final confidence score using the configured weighting.

6. Weight Configuration

The configured signal weights are:

- Hesitation: 20%
- Response Length: 15%
- Pace: 15%
- Sentiment: 15%
- Uncertainty: 20%
- Consistency: 15%

The weights are used to calculate the final confidence score.

7. Confidence Score

The engine produces a confidence score within the configured range:

Minimum: 0

Maximum: 100

The individual communication signals are combined using their configured weights to produce the final score.

8. Confidence Categories

The configuration defines the following confidence thresholds:

Strong:

75 and above

Moderate:

50 to 74

Lower confidence:

Below 50

These categories describe observable communication signals and should not be interpreted as psychological or personality assessments.

9. Structured Output

The analysis produces structured information containing the relevant communication signals and calculated confidence information.

The output can be used by downstream components of the screening workflow.

The analysis can include:

- Confidence score
- Confidence category
- Hesitation signal
- Response length signal
- Pace signal
- Sentiment signal
- Uncertainty signal
- Consistency signal

10. Implementation

Main implementation:

scoring/confidence_sentiment_engine.py

Configuration:

data/confidence_sentiment_configuration.json

Automated tests:

tests/test_confidence_sentiment_engine.py

11. Testing

The Day 27 automated test suite validates:

- Empty response handling
- Hesitation detection
- Uncertainty detection
- Positive sentiment detection
- Negative sentiment detection
- Response length signal
- Confidence score range
- Confidence category
- Consistency signal
- Structured output

Test Result

All Day 27 Confidence and Sentiment Analysis tests passed successfully.

12 passed in 0.09s

12. Integration

The Confidence and Sentiment Analysis Engine provides an additional communication-signal analysis layer for the AI screening workflow.

It can operate on candidate responses generated through the existing screening and transcript-processing components.

The resulting structured information can be used together with other screening signals for downstream candidate evaluation.

13. Day 27 Deliverable

The completed Confidence and Sentiment Analysis Engine provides a configurable framework for analyzing observable communication signals in candidate responses.

The system can detect hesitation, uncertainty, sentiment, response characteristics, and consistency indicators and combine these signals into a structured confidence analysis.

The implementation is supported by automated tests and uses configuration-driven scoring weights and thresholds.